

Minghao (Mark) Liu

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Education

- Hong Kong University of Science and Technology (HKUST)** 2026-
PhD candidate in Computer Science
- Hong Kong University of Science and Technology (HKUST)** 2022 – 2026
BEng in Computer Science | Minor in Mathematics
- Washington University in St. Louis** Fall 2024
Exchange Student, McKelvey School of Engineering

Research Interests

- **Deep Research Agents.** Constructing effective **experience trajectories** that efficiently fuse **internal knowledge** with **web-integrated information**, enabling stronger multi-step reasoning.
- **Auditable AI Systems.** Designing **explainable, systematic benchmarking** frameworks that diagnose failure patterns in AI models, informing improvements in efficiency and reliability.

Publications

- **LeanForPhysics: Comprehensive Reasoning Framework for University-level Physics in Lean4**
Yuxin Li*, **Minghao Liu***, Ruida Wang*[†], Ji Wenzhao, Zhitao He, Rui Pan, Junming Huang, Tong Zhang, Yi R. Fung
ICLR 2026, *Equal contribution
- **MedEBench: Diagnosing Reliability in Text-Guided Medical Image Editing**
Minghao Liu, Zhitao He, Zhiyuan Fan, Qingyun Wang, Yi R. Fung
Findings of EMNLP 2025
- **Scaling Environments for LLM Agents: Fundamentals, Approaches, and Future Directions**
Yuchen Huang, Sijia Li, Zhiyuan Fan, **Minghao Liu**, Wei Liu, Yi R. Fung
SEA @ NeurIPS 2025
- **A Benchmark for Evaluating Purchase Intention Comprehension Abilities of Large Language Models in E-commerce**
Wenxuan Ding*, Weiqi Wang*, Sze Heng Douglas Kwok, **Minghao Liu**, Tianqing Fang, Jiaxin Bai, Xin Liu, Changlong Yu, Zheng Li, Chen Luo, Qingyu Yin, Bing Yin, Junxian He, Yangqiu Song
Findings of EMNLP 2024, *Equal contribution

Awards & Scholarships

- Talent Development Scholarship – HKSAR Government Scholarship Fund 2023
- University's Scholarship Scheme for Continuing Undergraduate Students 2023–24
- HKUST Alumni Endowment Fund High Flyers Program Scholarship 2023–24
- HKSAR Government Scholarship Fund – Reaching Out Award 2024–25
- Dean's List 2022–25
- Hong Kong PhD Fellowship Scheme 2026
- Red Bird Phd Award 2026